

Eliminating the Observer Effect: Wave Function Collapse as Deterministic Topological Reconnection in a Condensate Vacuum

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Repository: <https://github.com/infosave2007/vmf>

Abstract

The “measurement problem” and the subsequent “collapse of the wave function” remain the most philosophically contentious aspects of the Copenhagen interpretation of quantum mechanics, notoriously requiring an active, conscious observer to define reality. We propose a deterministic, fluid-dynamic resolution derived from the Null-Vector Gravity (NVG) framework, wherein the quantum vacuum is treated not as empty space, but as a macroscopic, non-perturbative Quantum Chromodynamics (QCD) condensate. In this model, an elementary particle (such as an electron) is not a point-mass governed by abstract probability amplitudes, but a stable topological defect—a localized displacement within the vacuum conformal structure. Prior to interaction, this defect propagates as a spatially extended solitary wave. We demonstrate mathematically that what is conventionally termed “wave function collapse” is the onset of a purely deterministic topological reconnection event. When the extended wave envelope encounters a macroscopic density gradient (e.g., the lattice of a detector screen), the vacuum displacement vectors undergo rapid non-linear contraction, “snapping” the delocalized energy back into a localized, point-like topological knot. This fluid-dynamic mechanism entirely eliminates the need for wave-particle duality and the observer effect, restoring objective realism to quantum mechanics while fully preserving the empirical interference patterns predicted by the Schrödinger equation.

1. Introduction

Since the formalization of quantum mechanics in the 1920s, the Copenhagen interpretation has mandated a profound departure from objective realism. Central to this paradigm is the Born rule, which posits that the wave function $\psi(x)$ represents merely a probability amplitude. According to standard theory, a particle exists in a state of ubiquitous superposition, possessing no definite position or momentum until the exact moment of “measurement” or “observation,” at

which point the extended wave function instantaneously collapses into a Dirac delta function.

This paradigm introduces severe ontological paradoxes—most famously illustrated by Schrödinger’s cat—and requires a non-local, instantaneous physical alteration across space that deeply disturbed Einstein, Podolsky, and Rosen (EPR). While alternative frameworks such as De Broglie-Bohm pilot-wave theory and spontaneous objective collapse models (e.g., GRW theory, Penrose’s gravitationally induced collapse) attempt to restore determinism, they often require the introduction of unobservable hidden variables or ad-hoc modifications to the Schrödinger equation.

In this letter, we introduce a fundamentally different approach utilizing the Null-Vector Gravity (NVG) framework. By upgrading the ontological status of the vacuum to a structured physical fluid (anchored by the QCD mass gap), we reinterpret the wave function not as probability, but as a literal physical strain in the vacuum fabric.

2. The Vacuum as a Macroscopic Condensate

The core premise of the NVG framework is that the mass of fundamental particles is an effective coupling to a ubiquitous vacuum field, scaling with the non-perturbative QCD anchor $M_{\Omega,0} = 859$ MeV. The vacuum is a dense, highly structured condensate.

In this framework, an elementary particle is not an infinitely small point metric. It is a stable topological defect—a localized “knot” or vortex—within the continuous vector field of the vacuum. Because the vacuum is an elastic medium, the energetic presence of this defect induces a strain that extends indefinitely outward.

Therefore, a traveling electron is a physical soliton. The core of the defect carries the quantized charge and the bulk of the rest mass, while the surrounding “wake” in the vacuum fluid constitutes the spatially extended wave.

3. The Double-Slit Experiment Reinterpreted

When a single electron is fired towards a double-slit apparatus, the Copenhagen interpretation asserts that the point-particle enters a superposition of states, traversing both slits simultaneously as a probability wave, interfering with itself, and only deciding where to land upon striking the detector.

In the NVG fluid-dynamic model, the process is entirely classical and deterministic. The electron is a physical wave of vacuum displacement. As this spatially extended wave packet approaches the barrier, the broad “wake” physically passes through both slits simultaneously, creating two secondary wavefronts of vacuum strain that undergo standard physical interference, exactly analogous to ripples in a fluid surface.

The core topological defect (the localized knot of mass-energy) passes through only one of the slits, guided by the local gradients of the total interference pattern generated by its own extended wake. The electron is “riding” its own self-generated standing wave in the vacuum condensate.

4. Deterministic Topological Reconnection (Collapse)

The defining moment of quantum mechanics is the registration of the particle on the detector screen as a single, highly localized point. Why does the extended interference wave “collapse”?

In the NVG framework, the detector screen is a dense macroscopic lattice of other stable topological defects (atoms). When the diffuse, extended wavefront of the electron’s vacuum strain encounters this dense barrier, it experiences a drastic impedance mismatch. The continuous conformal field lines of the vacuum strain are violently interrupted.

This triggers a well-known phenomenon in fluid dynamics and plasma physics: **Topological Reconnection**. The vacuum vector field must rapidly reconfigure to minimize energy. The non-linear tension in the vacuum fabric causes the extended wave envelope to rapidly contract, “snapping” the delocalized strain energy back toward the core topological defect.

The wave does not vanish probabilistically; it retracts physically and deterministically into a localized point-knot. The location of this final “snap” is deterministically governed by the peaks and troughs of the physical interference pattern that arrived at the screen moments prior.

5. Conclusion

The Null-Vector Gravity framework provides a robust physical ontology for quantum mechanics that completely eliminates the necessity of the “observer.”

1. **Wave-Particle Duality is an Illusion:** There is no duality. Particles are permanent topological defects in a fluid vacuum. They always possess a core (particle-like behavior) and an extended strain field (wave-like behavior).
2. **The Wave Function is Physical:** The Schrödinger equation does not calculate probability; it calculates the deterministic evolution of the physical vacuum strain density.
3. **Collapse is Reconnection:** “Measurement” is simply a physical collision between an extended vacuum strain wave and a dense atomic lattice, triggering a rapid, non-linear topological reconnection of the vacuum field lines.

By anchoring quantum mechanics in the hydrodynamics of the non-perturbative QCD vacuum, we restore objective realism to the microscopic universe. The moon is there whether we look at it or not, and the electron collapses deterministically whether an observer is conscious of it or not.

Data Availability

Interactive 3D visualizations demonstrating the continuous propagation and deterministic topological reconnection of the NVG wave-particle mechanism are available in the public repository at: [<https://github.com/infosave2007/vmf>].

References

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